كلية الهندسة
جامعة المنصورة

برنامج هندسة
الإلكترونيات والاتصالات

THE

OCTACORE

PROJECT

BCD TO EXCESS-3

CONVERTER

أعضاء الفريق

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Contents

1 BCD and Excess-3 definitions

1.1 BCD (Binary Coded Decimal)

- **Definition:** Binary Coded Decimal (BCD) is a class of digital encodings where each digit of a decimal number is represented by a fixed number of bits, usually four. The most common format is the 8421 BCD code, where the numbers represent the weights of the four bits $(2^3, 2^2, 2^1, 2^0)$.
- **Key Characteristics:**
  - Weighted Code: Each bit position has a specific mathematical weight.
  - Valid States: It only uses 10 valid combinations (from 0000 to 1001) to represent the decimal digits 0 through 9.
  - Invalid States: The remaining six combinations (1010 to 1111) are invalid in BCD and are treated as ”Don't Care” conditions in logic circuit design.

1.2 Excess-3 (XS-3) Code

- **Definition:** Excess-3 is a non-weighted digital code used to express decimal numbers. It is derived directly from the standard BCD code by adding the binary value of 3 (0011) to each BCD digit representation.
- **Key Characteristics:**
  - Non-Weighted Code: The bit positions do not carry specific geometric or arithmetic weights.
  - Self-Complementing Property: This is its most important feature. The 1's complement of an Excess-3 number is exactly equal to the Excess-3 representation of the 9's complement of the original decimal digit.
  - Application: This self-complementing nature significantly simplifies the hardware design for subtraction operations in Digital Systems and Arithmetic Logic Units (ALUs).

2 Truth Table and Function modeling

2.1 Truth Table

this is the Truth Table of converting BCD to Excess-3 As observed, all combinations beyond 9 are

Table 1: Full Truth Table for BCD to Excess-3 Code Converter

Decimal	**BCD Inputs**				**Excess-3 Outputs**			
Digit	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0
10	1	0	1	0	X	X	X	X
11	1	0	1	1	X	X	X	X
12	1	1	0	0	X	X	X	X
13	1	1	0	1	X	X	X	X
14	1	1	1	0	X	X	X	X
15	1	1	1	1	X	X	X	X

treated as 'Don't Care' conditions (X). This is because the BCD system is strictly designed to represent decimal digits from 0 to 9.

2.2 karnaugh-map

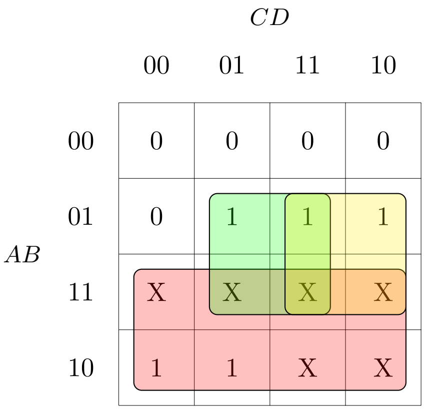

Figure 1: K-Map for Output
$W = A + BC + BD$

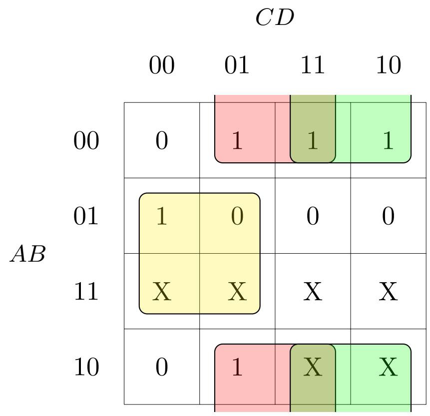

Figure 2: K-Map for Output
$X = \bar{B}C + \bar{B}D + B\bar{C}\bar{D}$

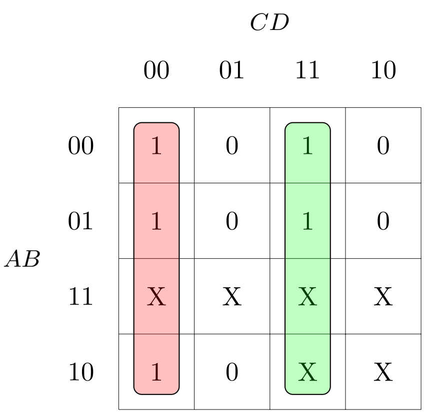

Figure 3: K-Map for Output
$Y = \bar{C}\bar{D} + CD$

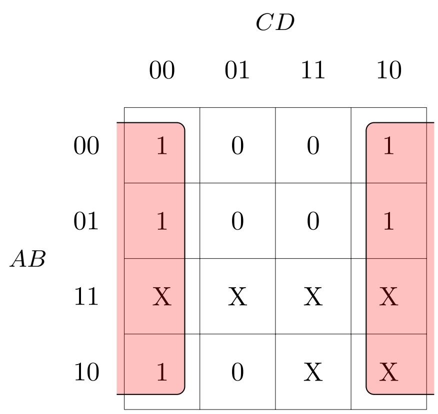

Figure 4: K-Map for Output
$Z = \bar{D}$

3 Logic Circuit Implementation and Modeling

In this section, the simplified Boolean expressions obtained from the Karnaugh maps are translated into practical digital logic circuits. The implementation details the hardware components required, individual circuit descriptions for each output, and finally, the unified combined system.

3.1 Required Hardware Components (TTL ICs)

To implement the system practically on a breadboard, standard Transistor-Transistor Logic (TTL) Integrated Circuits (ICs) from the 74xx series are utilized:

- **IC 7404 (Hex Inverter):** Contains 6 NOT gates, used to generate the complemented literals ($\bar{B}$, $\bar{C}$, and $\bar{D}$).
- **IC 7408 (Quad 2-Input AND Gate):** Contains 4 AND gates, used to generate product terms such as BC, BD, $\bar{B}C$, $\bar{B}D$, CD, and $\bar{C}\bar{D}$.
- **IC 7432 (Quad 2-Input OR Gate):** Contains 4 OR gates, used to sum the product terms and generate the final outputs.

3.2 Individual Logic Circuit Modeling

3.2.1 Circuit for Output W ($W = A + BC + BD$)

The Most Significant Bit (MSB) output W is implemented using:

- Two 2-input AND gates to realize the product terms BC and BD.
- Two cascaded 2-input OR gates to combine input A with the outputs of the AND gates, resulting in the final expression $W = A + BC + BD$.

3.2.2 Circuit for Output X ($X = \bar{B}C + \bar{B}D + B\bar{C}\bar{D}$)

Since the implementation strictly avoids XOR gates, the standard Sum-of-Products (SOP) structure for X requires:

- Three NOT gates to provide the inverted signals $\bar{B}$, $\bar{C}$, and $\bar{D}$.
- Three AND configurations: two 2-input AND gates for $\bar{B}C$ and $\bar{B}D$, and two cascaded 2-input AND gates to construct the 3-input product term $B\bar{C}\bar{D}$.
- Two cascaded 2-input OR gates to aggregate the three product terms into the output X.

3.2.3 Circuit for Output Y ($Y = \bar{C}\bar{D} + CD$)

The circuit for output Y is straightforward and symmetrical, utilizing:

- Two NOT gates to obtain $\bar{C}$ and $\bar{D}$.
- Two 2-input AND gates to compute the terms $\bar{C}\bar{D}$ and CD simultaneously.
- One 2-input OR gate to add both terms together, yielding $Y = \bar{C}\bar{D} + CD$.

3.2.4 Circuit for Output Z ($Z = \bar{D}$)

The Least Significant Bit (LSB) output Z is the simplest function in the converter. It requires no combinational logic combinations, only a single NOT gate connected directly to the input line D to output its complement ($\bar{D}$).

3.3 Combined Integrated Circuit Design

To construct the complete BCD to Excess-3 code converter as a single operational module, all four individual sub-circuits are integrated on the breadboard layout.

A shared **Input Rail Bus** is created consisting of seven vertical tracks: four for the primary inputs (A, B, C, D) and three for the inverted outputs ($\bar{B}, \bar{C}, \bar{D}$) emerging from the IC 7404. This central bus distributes the signals efficiently across the AND gates (IC 7408) and OR gates (IC 7432) arrays. Integrating the functions into a single system minimizes the total wiring clutter, drastically reduces propagation delay, and ensures that all four outputs (W, X, Y, Z) update simultaneously in real-time as the 4-bit BCD input changes.

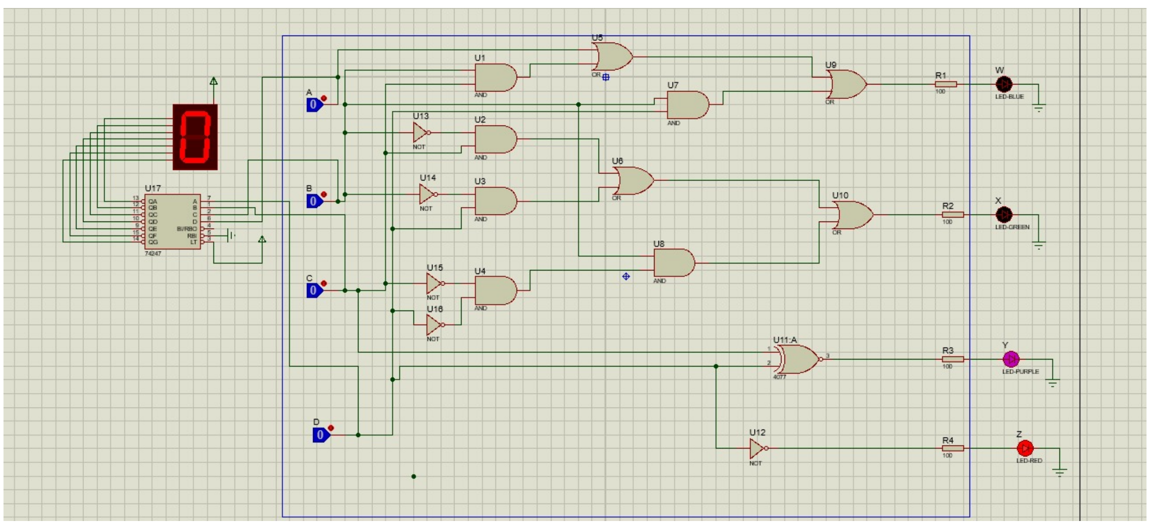

Figure 5: Combined Logic Circuit Design

4 Display Interface and BCD-to-7-Segment Decoder Integration

To provide a clear, human-readable visual verification of the 4-bit BCD input, a 7-segment display interface was integrated into the system. This allows the user to immediately observe the decimal equivalent of the binary input being processed by the converter.

4.1 Decoder IC Selection and Functionality

Since a 4-bit binary code cannot drive a 7-segment display directly, the dedicated **SN74LS47N** BCD-to-7-segment decoder/driver IC is utilized as the intermediary hardware driver:

- **IC Specifications:** The SN74LS47N is a Low-power Schottky (LS) digital logic device housed in a standard 16-pin Dual In-line Package (DIP-16).
- **Decoding Logic and Outputs:** The IC accepts the 4-bit BCD input lines (A, B, C, D) and decodes them into a 7-bit layout to drive the display segments (a through g). It features active-low, open-collector outputs capable of sinking the necessary current to illuminate the segments.
- **Display Compatibility:** Due to its active-low output configuration, the SN74LS47N is strictly compatible with a **Common Anode 7-segment display**, where the common pin of the display is tied directly to the positive 5V VCC power rail.

4.2 7-Segment Display Wiring and Protection

The Common Anode 7-segment display consists of seven directional LEDs sharing a single positive terminal, which are selectively grounded by the decoder to form numerical digits from 0 to 9.

- **Current Limiting Resistors:** To protect the individual LED segments of the display from overcurrent damage and to prevent excessive power consumption, a resistor network consisting of seven 220 Ω (or 330 Ω) resistors was placed in series between the SN74LS47N active-low output pins and the display inputs.
- **System Operation:** When a valid BCD input is applied to the system, the SN74LS47N instantaneously maps the binary state and grounds the corresponding segment pins, delivering a real-time decimal readout that synchronizes perfectly with the conversion process.

5 Printed Circuit Board (PCB) Design and Fabrication

To transition the BCD to Excess-3 code converter from a temporary breadboard prototype to a permanent, reliable, and robust hardware system, a Printed Circuit Board (PCB) was designed. This section outlines the structural phases of the PCB development workflow, from schematic capture to physical manufacturing preparation.

5.1 Schematic Capture and Component Selection

The finalized logic circuit configuration was translated into a professional PCB computer-aided design (CAD) suite (EasyEDA).

– **Component Footprints:** Standard DIP (Dual In-line Package) footprints were chosen for all ICs (DIP-14 for the 7404, 7408, and 7432 chips) to allow the integration of IC sockets, ensuring easy component testing and replacement.

– **Connectors and Interface:** Screw terminal blocks or male header pins were allocated for the 4-bit BCD input (A, B, C, D), the 4-bit Excess-3 output (W, X, Y, Z), and the 5V VCC/GND DC power input rails.

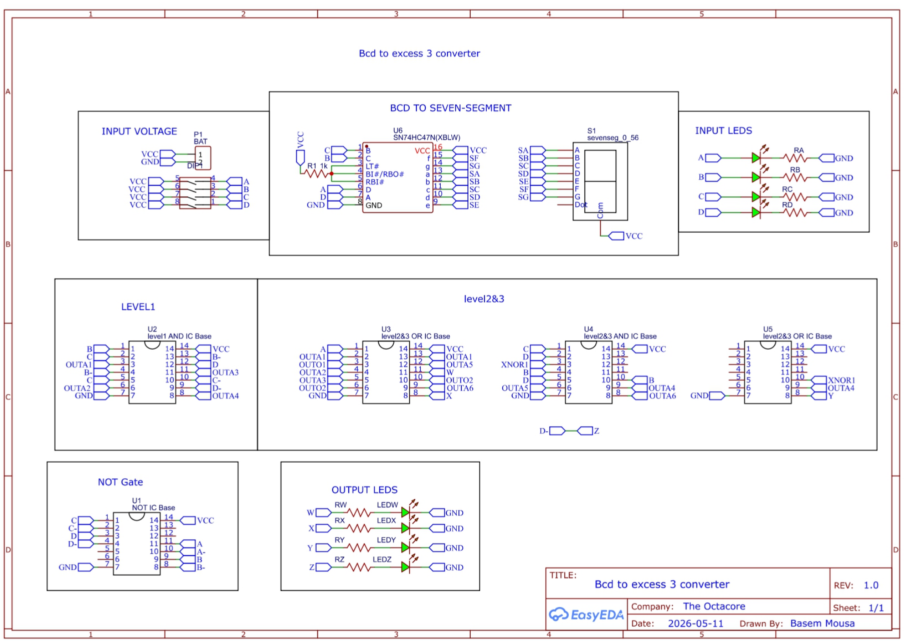

Figure 6: Schematic

5.2 PCB Layout and Routing Strategy

Following the logical design, the components were transferred onto a physical board layout environment where spatial constraints and signal integrity rules were strictly applied:

- **Board Geometry:** A compact double-layer (Top and Bottom copper layers) board profile was selected to balance structural ergonomics and minimize production cost.
- **Component Placement:** The IC sockets were aligned symmetrically in a single row to optimize trace length, maintaining short and direct paths from the input headers to the logic gates.
- **Trace Width and Planes:** A uniform trace width of 0.5 mm was applied to all power and signal tracks across the board. This unified width guarantees an optimal current-carrying capacity for both the VCC and GND lines while minimizing trace resistance across all logical paths. Since the circuit operates at standard low-frequency digital logic, a copper pour was omitted, and standard routed ground traces were utilized to complete the circuit loops.

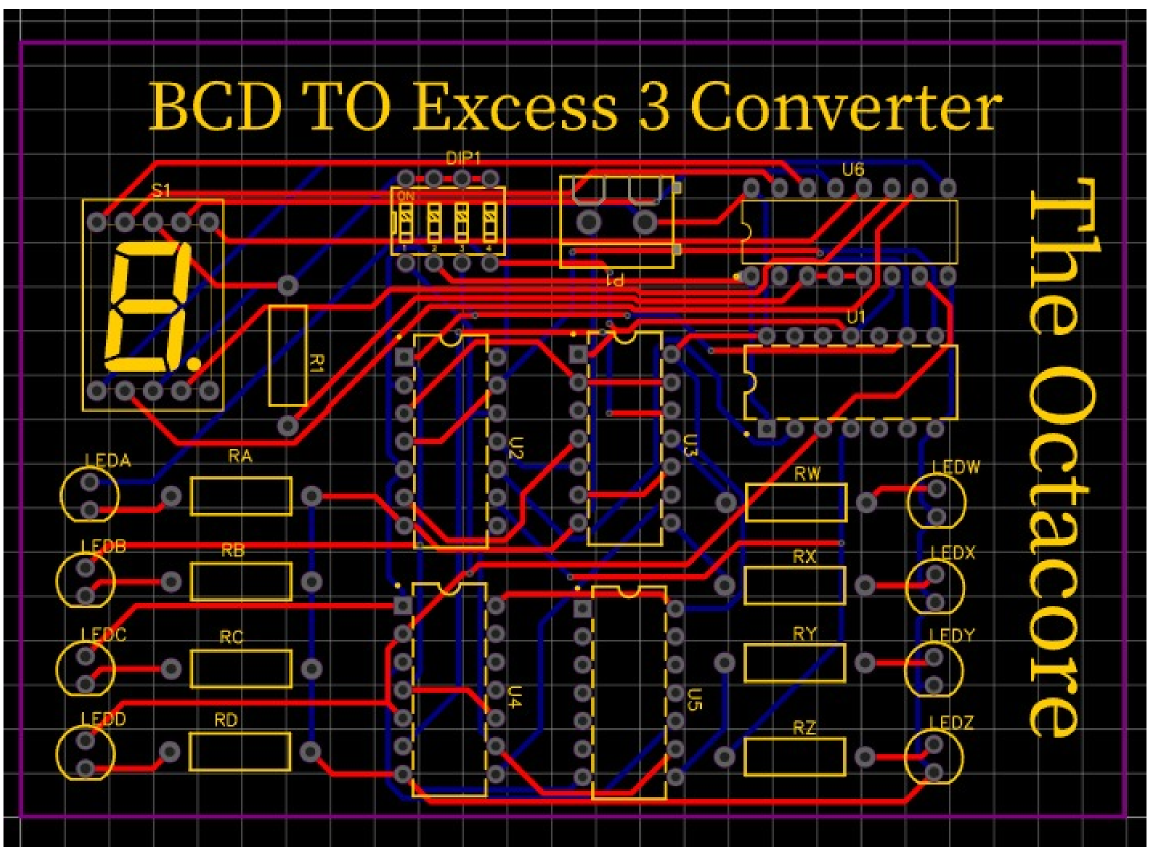

Figure 7: PCB Layout

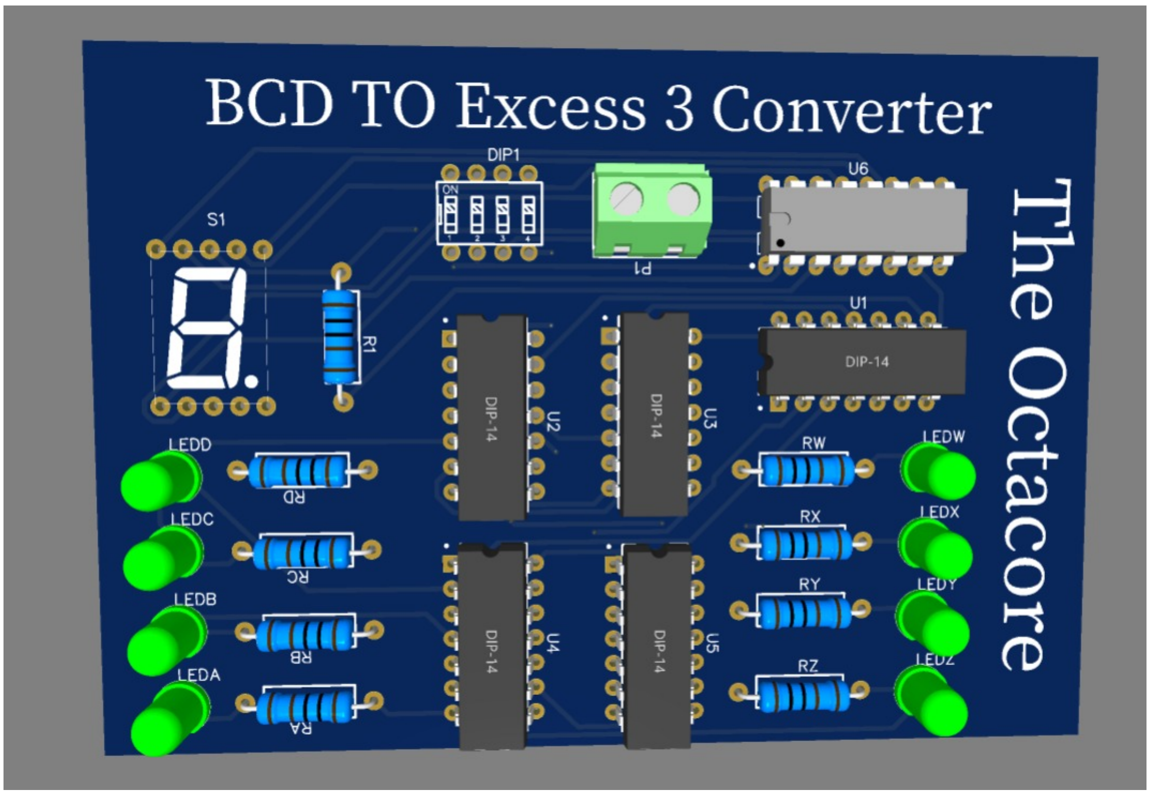

Figure 8: 3D Layout